BizRadar AI — Technical Guide

Know your market in about a minute. Sourced research by a team of AI agents.

1. What BizRadar AI does

A small founder types a **business idea, a city and a target customer**. In about a minute, a team of AI agents researches the local market and writes a sourced report:

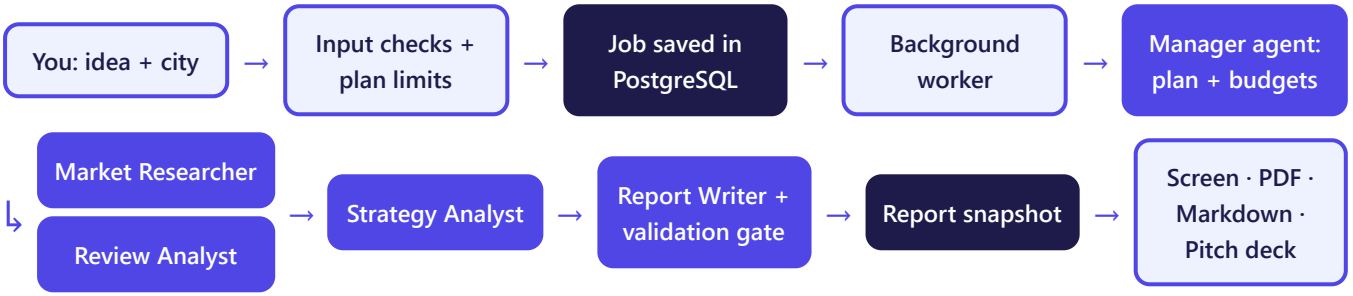
You give	You get
Business idea, city, target customer	Real competitors with prices, strengths and weak spots
(optional) competitors you already know	Customer complaints and needs from reviews, maps, apps, social media and forums
Which platforms to search	Market gaps rated by confidence, and a suggested price
	A 30-day launch plan and a first ad message
	Downloads: Markdown, PDF and a 6-slide PowerPoint pitch deck
	Weekly monitoring with alerts when prices or competitors change

What makes it special

- **Every claim has a source.** Each line carries a tag like `S34` that points to a page the system actually saved. Anything the AI writes without a saved source is **removed** before the report is built.
- **Honest labels.** Claims are marked **Fact**, **Estimate**, **Inference** or **Unknown**.
- **Agents you can watch.** The research screen shows each agent working live, using real progress saved in the database, never a fake animation.
- **Built for India.** It searches Google, Google Maps, Justdial, Sulekha, IndiaMART, magicpin, Swiggy, Zomato, Instagram, Facebook, YouTube, Reddit, Quora, Trustpilot and MouthShut, and uses ₹ prices.
- **Never overspends.** Every run has hard limits on AI tokens, pages, searches and time. When a limit is hit, the run stops safely and is marked **Partial**, never padded with invented content.

One real sample run (kids AI coding classes, Coimbatore): **79 sources, 6 platforms, 13 searches, 3 competitors, 9 report sections, about 15k tokens.**

2. Journey of a research request



Filled blue = AI agents · Navy = saved in the database · The two researchers run at the same time.

1. **You submit the form.** Inputs are cleaned and checked (length, format, allowed platforms), and your plan's limits are checked on the server.
2. **The job is saved and the page returns at once.** Research never runs inside the web page, so a refresh or a closed tab never loses the work.
3. **A background worker claims the job** atomically, so two workers can never take the same one. It sends a heartbeat while it works.
4. **The Manager plans the run** inside its token, page, search and time budgets.
5. **Market Researcher and Review Analyst run at the same time** (two parallel workers): one finds competitors, prices and offers; the other finds customer complaints and needs.
6. **The Strategy Analyst** turns only validated findings into gaps, positioning, a suggested price, tactics and a 30-day plan.
7. **The Report Writer** assembles the report and runs the **validation gate**: every claim and price must point to a saved source, or it is dropped.
8. **The report is saved once as a snapshot.** The screen, PDF, Markdown and pitch deck all read the same snapshot, so they always agree.

3. The AI agent team

Agent	Job	Key safety rule
Manager	Plans the searches and watches the budgets	Stops the run when a limit is reached
Market Researcher	Finds real competitors, websites, prices, offers, ratings	A competitor with no saved source is dropped
Review Analyst	Groups customer complaints and needs into themes	Themes must cite saved review pages
Strategy Analyst	Gaps, SWOT, positioning, suggested price, tactics, 30-day plan	Each tactic must target a real complaint theme
Report Writer	Builds the report and checks every claim	Unsourced prices and claims are removed
Pitch Deck agent	Six editable PowerPoint slides	Uses **no AI call**: built only from the saved report

How the agents are built. A small custom orchestrator (`agents/pipeline.py`) follows the hierarchical "manager + specialists" pattern. Each agent is one OpenAI call that must return a **typed Pydantic form** (`agents/schemas.py`) through OpenAI structured outputs, so the app gets clean data, not loose text. Every step writes real progress events to the database, and the research screen reads them to show the agents live.

4. Technology used and why

Part	Technology	Why it was chosen
Web app	**Python + Streamlit**	Fast to build rich data screens in one language
Server routes	**Starlette (ASGI)**	Secure cookies, email links and Google sign-in alongside Streamlit

Part	Technology	Why it was chosen
Database	**PostgreSQL 16 + SQLAlchemy 2 + Alembic**	Reliable data, versioned schema changes (8 migrations)
Job queue	**Database-backed queue + worker**	Durable jobs that survive restarts, without an extra service
AI	**OpenAI GPT-4o-mini**, structured outputs	Low cost per run, typed answers; data not stored by OpenAI (`store=False`)
Search	**Serper** (Google Search + Google Places)	Real Google results, ratings and review counts for India
Page reading	**Guarded fetcher** (httpx)	Reads public pages safely (see section 6)
Login	**bcrypt**, server-side sessions, **Google OAuth**	Strong password storage, revocable sessions
Exports	**fpdf2** (PDF), **python-pptx** (pitch deck)	PDF with ₹ support and fully editable slides
Hosting	**Docker Compose** on a Hostinger VPS, **nginx**, **Let's Encrypt**	Same setup everywhere, HTTPS, auto-restart
Testing	**pytest** (110 tests), **ruff**	Tests never call paid APIs

The AI, search, page-fetch and email services each sit behind a small interface in `providers/`, so any of them can be swapped (for example a different AI model or email service) without touching the agents.

5. Features covered

- **Landing page** with a read-only **sample account** and **demo videos**.
- **Sign up, email verification, log in, Google sign-in, forgot and reset password, change password.**
- **New research** form, which turns into the **live agent view** when you press Start (agent network, timeline, findings and source coverage as they happen). Runs can be **cancelled**.
- **Report**: executive summary, competitors, pricing chart with the suggested price, complaint themes, gaps, SWOT, tactics, 30-day plan, sources and limitations.
- **Downloads**: Markdown, PDF and a **6-slide pitch deck**.
- **History and comparison** of two reports; **re-run** a project.
- **Dashboard** as the home screen for returning users.
- **Weekly monitoring**: scheduled re-research, a change feed, and alerts when prices or competitors change.
- **Plans and billing** (Free, Starter, Pro) in test mode, with ₹0 test invoices.
- **Settings**: profile, password, notifications, projects and **Delete my data**.
- **"Was this useful?"** feedback on each report.

6. Security and agent safety

Accounts and data

- Passwords hashed with **bcrypt** (minimum 10 characters). A dummy hash check hides whether an email exists.
- Sessions use an **HttpOnly cookie**; reset, verification and session tokens are stored only as **SHA-256 hashes**. Sessions last 12 hours (30 days with "remember me").
- **5 failed logins lock the account for 15 minutes**; signup, login, reset, contact and feedback are **rate-limited**.
- Reset links expire in **30 minutes**, verification links in **24 hours**.
- Google sign-in uses a signed, short-lived **state cookie** against forged logins.
- **Every query is scoped to its owner.** Someone else's report and a missing report get the same "not found" answer, so nothing leaks.
- An **audit log** records logins, research runs, plan changes, feedback and deletions.
- Security headers (`nosniff`, `X-Frame-Options: DENY`); HTTPS through nginx; secrets only in `.env`.
- The public sample account is **read-only on the server**. A private-testing switch can limit login to chosen emails.
- Billing collects **no card data**.

Agent safety

- **Prompt-injection defence.** All web text and user input is wrapped as `<untrusted_data>` and kept apart from the agent instructions. Agents are told to ignore any instructions inside it and never reveal their own instructions (`agents/prompts.py`).
- **Restricted tools.** Agents cannot browse freely; they only get search results and pages that the guarded fetcher has already read.
- **Guarded fetcher** (`providers/fetch/guarded.py`): only public `http/https` sites on standard ports; private, local and reserved addresses blocked (stops SSRF attacks); every redirect re-checked (max 3); **robots.txt respected**; 10-second and 1.5 MB limits. Blocked pages are shown honestly in the report.
- Report text is cleaned so web content cannot inject links or HTML into exports.

7. Truth checks and data quality

- **Grounding check.** Every source reference the AI returns is checked against the pages saved in this run. A competitor, theme, gap or SWOT point without a valid source is **dropped**.
- **Prices must have a source.** A price without one is removed, and the report says so.
- **Tactics must be earned.** A "how to win their customers" tactic is kept only if it targets a real complaint theme with evidence.
- **Matching map listings** to competitors needs a strong name match (60%+), so ratings are not attached to the wrong business.
- **Your own idea** has no reviews yet, so its SWOT is shown as **Inference**, at most 3 points each.
- **Validation gate.** The report is marked **Partial** (with the reasons listed) when there are fewer than 3 verified competitors, no strategy could be built, no tactic had enough evidence, or a limit stopped the run.
- A run that finds **no competitor does not count** against your plan.
- A **methodology** box shows sources found, pages read, pages blocked, searches and tokens used.

8. Reliability and error handling

Problem	What BizRadar does
Worker crashes mid-run	Heartbeat stops; after 120 s the run is **retried once** from a clean start, then marked failed
AI call fails or is slow	60 s timeout and 2 automatic retries
AI returns broken output	Structured output is required; a missing form is treated as an error, never guessed
Search or page fails	12 s / 10 s timeouts; the page is marked failed or blocked and research continues
Run takes too long	**10-minute wall-time limit**; the report is saved as Partial
You cancel	Checked between every step; open agents are closed as "skipped"
Weekly check runs twice	A unique slot key per schedule time means **no duplicate runs**; alerts are deduplicated
Server restarts	Containers restart automatically; jobs live in the database, so nothing is lost

9. Cost and speed controls

Control	Free	Starter / Pro
Reports	1 (lifetime)	5 / 20 per month
Competitors per report	3	5
AI token cap per run	60,000	250,000
Pages fetched per run	25	80
Monitored projects	0	2 / 10

- Searches per run are capped (8 + 4 per competitor), and the run stops at **92% of its token budget**, so it never goes over.
- Limits are enforced **on the server**, never only in the screen.
- The two specialist agents run **in parallel** to save time.
- **Validation mode** replays saved real runs for testers with **zero** OpenAI or Serper calls, and live runs are capped per day across all accounts.
- The pitch deck uses no AI call at all.

10. Monitoring, testing and deployment

- **Run tracking:** status, stage, tokens, pages, searches, attempts, heartbeat and error are saved for every run.
- **Logging** in the app and worker; Docker logs are size-capped (10 MB × 3 files).
- **Audit log** of important actions in the database.
- **110 automated tests** (pytest) covering login, research, reports, monitoring, billing, pitch deck, validation and pages. Tests use fake providers and **never call paid APIs**. Code style checked with ruff.

- **Deployment:** Docker Compose (app, worker, PostgreSQL) on a Hostinger VPS, behind nginx with a Let's Encrypt HTTPS certificate. `deploy/deploy.sh` pulls the latest code from GitHub and rebuilds.
- **Backups:** a nightly PostgreSQL backup at 02:30, keeping the last 14 days (`deploy/backup.sh`).

11. Code map

Folder	What's inside
`agents/`	Research pipeline (Manager + specialists), prompts, typed schemas, sample replay
`core/`	Services: auth, research, reports, exports, pitch deck, monitoring, billing, plans, validation
`providers/`	Adapters for OpenAI, Serper search, guarded page fetcher, email
`worker/`	Background job runner and the weekly monitoring tick
`app/` , `app_pages/`	Streamlit components, styles and the screens
`db/migrations/`	Alembic database migrations
`deploy/`	Production Docker Compose, nginx config, deploy and backup scripts
`tests/`	pytest suite

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